Seokhyeon Lee

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Research Interests: HW-Aware ML Algorithms, Algorithm-HW Co-design, Systems for AI, On-device AI
Skills: C/C++, Python, CUDA, CUTLASS, OpenMP, PyTorch, TensorFlow, Git, GitHub Actions (CI)

EDUCATION

Sungkyunkwan University — South Korea
B.S. in Computer Science and Engineering — Mar. 2023 – Feb. 2025
Chonnam National University — South Korea
B.S. Candidate in Mechanical Design Engineering (Transferred to SKKU in 2023) — Mar. 2017 – Feb. 2023

WORK EXPERIENCE

MOREH — South Korea
NPU Software Engineer — May 2025 – Jun. 2026

- Implemented and optimized Tenstorrent NPU kernels for distributed LLM training and inference across pytest validation, Python/C++ bindings, host runtime, device kernels, and the SFPU's SIMD instructions
- Designed and implemented a multicore cumsum kernel based on the Sklansky parallel-prefix sum, using NoC multicast and inter/intra-core semaphore synchronization, achieving a 110× speedup over the baseline tt-metal kernel
- Diagnosed and fixed a nondeterministic hang in composite collective-communication kernels caused by trace replay corrupting L1-resident semaphores (analogous to CUDA Graph replay) #40276
- Developed a vLLM plugin for Tenstorrent NPUs, enabling end-to-end serving of models from a few billion to hundreds of billions of parameters (e.g., Qwen3, Llama3, GPT-OSS), and resolving stability and performance regressions #43692

PERSONAL PROJECT

HPMC: High-Performance MNIST Classification — Feb. 2025 – Mar. 2025

- Built a deterministic end-to-end MLP training pipeline in CUDA/C++ from scratch without cuBLAS or CUTLASS, achieving up to 8× training speedup and 13% lower GPU memory usage compared to the PyTorch baseline

AWARDS

Silver Prize, National Mathematics Competition for University Students — Korean Mathematical Society, 2023, 2024
Silver Prize, Algorithm Problem Solving Competition — Chonnam National University, 2021, 2022

ACTIVITIES

Sungkyunkwan University — South Korea

Undergraduate Research Intern at Dash Lab, supervised by Prof. Simon Woo — Jun. 2024 – Dec. 2024

- Read and presented research papers in the field of AI Security
- Contributed to a joint ML project with Kia Motors

Programming Tutoring

- Python Programming Tutor for Freshmen — 2024
- C++ Algorithm Tutor for CS Undergraduate Students — 2023